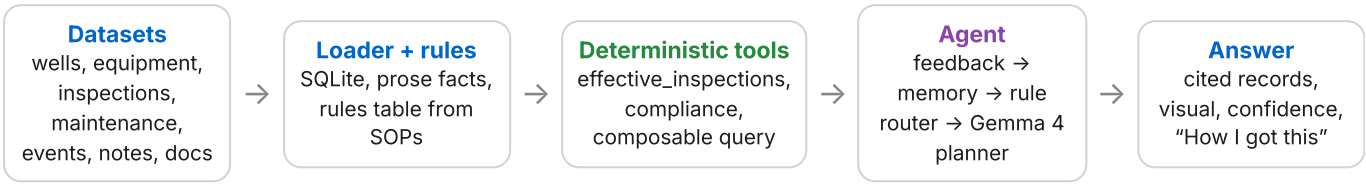
Asset Intelligence Copilot by Mangesh

Mangesh Raut · FlytBase challenge, AHC × Hacktoberfest 2026 · MIT

Repo: github.com/mangeshraut712/asset-intelligence-copilot · Video (4:41): vimeo.com/1236779581

System design

Principle: the model never does arithmetic. Every number comes from a deterministic, unit-tested tool over SQLite. Every threshold and deadline is extracted from the manuals and SOPs. Every answer cites record IDs and document passages. The open-weight **Gemma 4** model plans tool calls, phrases answers and reads photos, and is checked at each step.



Memory (scope + provenance) and the number guard sit around the agent; any LLM failure falls back to rules.

Storage and indexing

A re-runnable loader ingests any number of dataset folders and follows each `report_file` to extract facts that exist only in prose (*"contained about 7 hours after the inspection flight"*). It builds a `rules` table from the documents: producing at Δ ≥ 15 °C (T200) and 20 °C (T350); reportable at ≥ 5 bbl; notice within 24 h; containment within 4 h; verification within 60 d; NOV response within 30 d; RCA at 3 repairs. New wells, areas or manuals need no code change; 400 wells load in about 1 s.

Tools

`effective_inspections` is the single source of truth: raw readings + correction notes + runtime corrections, originals kept. Specialised tools cover status, temperature series, spills, downtime, compliance (SR-01/02/03, RC-02, RC-04, RCA), containment, reportable spills and repair history. A **composable query** covers the long tail: metric × scope (well, area, type, team, field) × window × group-by. For the finals, `copilot/final_tools.py` reads recall, regulator-letter, MOC and OPS-1/2/3/5 rules from the documents. No answers are hard-coded.

Agent (Gemma 4)

1) Feedback detection. 2) Private-memory recall. 3) A confident rule router. 4) Otherwise Gemma 4 as planner fills the structured query (function calling, JSON fallback). Arguments are validated; an invalid or empty result triggers one retry through the rule router. Summaries are accepted only if they add **no number absent from the tool output**. Gemma 4 also describes uploaded inspection photos, advisory only. Runs via Gemini API, OpenRouter or Ollama.

Memory and honesty

Level	Scope	Behaviour
Conversations	one user	resolves only from that user's chats
Well facts	all users	attributed, timestamped, confirmed / unconfirmed / planned
Corrections	one field	original + new value, author, reason; rules re-evaluated
Type rules	one well type	cites the user, not the manual

Contradicting claims are stored as *unconfirmed* and the conflicting inspection is named.

Demo layer, quality, cost

Proactive field briefing, clickable evidence drawer, "How I got this" trace, confidence and conflict badges, dark and light themes. 101 pytest tests, including the 18 examples (27/27) and 46 unseen questions computed independently from the raw files (46/46). Model cost: $0 (free Gemma 4). Runs locally; no hosted deployment.

Results on the 11 final questions

Organizers' test dataset (600 wells, commit 009e2b5), today = 15 Jan 2027. `AIC_TODAY=2027-01-15 python scripts/final_run.py --data <test-data>/dataset --only-test --questions eval/final_questions.json`. Every turn is answered by a deterministic tool; Gemma 4 isn't called on this set. Verdicts checked by hand against the raw files.

16/21	2	3	101	≈2.5 s
correct	mostly correct	judgement calls	tests passed	full run (was 185 s)

Turn	Question · answer (short)	Verdict
Q1	**Reservoir pressure map** Declines: no pressure field in inspections, maintenance or notes (only flowline test pressures in repair reports).	Correct
Q2	**Top oil producer 2026** Declines: no production volumes, only producing / not producing from thermal readings.	Correct
Q3	**Gas reports since 1 Nov** 26 reports: 17 met, 9 missed (24 h, 12 h from 1 Jan 2027). WELL-0332 missed (drone only).	Correct
Q4	**Flowline bypasses** WELL-0248 allowed (MOC-031 ext. by MOC-044 to 28 Feb 2027); WELL-0249 expired; WELL-0405 not approved. Past: 0087, 0158 expired; 0561 not approved.	Correct
Q5	**Regulator spill letter** 6 reportable spills in NE Block 7/8 since 1 Jun 2026; 3 not compliant (WELL-0123, 0182, 0262: no soil samples).	Correct
Q6	**Inspection vs note conflicts** 15 wells, applying OPS-3. Caveat: underload-restart notes (WELL-0219, 0249) counted as restarts.	Mostly correct
Q7.1	**Wells that sound wrong** Last-month abnormal notes, no repair since, ranked by a transparent (unvalidated) score.	Judgement call
Q7.2	**Top 3 crews this week** WELL-0182, WELL-0411, WELL-0297.	Judgement call
Q8.1	**WELL-0333 pump swapped** First-hand: counted as replaced, attributed to Nathan, no work order.	Correct
Q8.2	**Private sort preference** Saved as Nathan's private preference only.	Correct
Q8.3	**Recalled pumps in ground** 8 pumps. All 18 recalled-lot installs checked: 9 replaced by other lots, WELL-0333 excluded.	Correct
Q8.4	**Remove WELL-0577** Not removed: second-hand claim, no work order, flagged unconfirmed.	Correct
Q8.5	**My area + install cost** NW Block 12: WELL-0305, 0426, 0428; sorted by area then ID, costs in $K.	Correct
Q9.1	**Recalled pumps by area** 8: NE8 2, NW12 3, SE32 1, SW23 2. WELL-0577 listed as unconfirmed.	Correct
Q9.2	**Northwest Production lead** Nathan Pierce; team runs 150 wells.	Correct
Q9.3	**OPS-5 cold-weather check** OPS-5 quoted. Example: WELL-0310 (INS-0310-0213, 17 Dec, −6 °C, no heat-trace check in 7 days).	Correct
Q9.4	**H2S reading** No H2S reading recorded; only drone follow-ups, OPS-1 missed. Says so, no invented number.	Correct
Q9.5	**Producing status of the 8** WELL-0263 and WELL-0160 not producing; last repair shown for each.	Correct
Q9.6	**Most urgent well** WELL-0332: missed gas response (safety first), not producing.	Judgement call

Turn	Question · answer (short)	Verdict
Q10	**Delay causes 2026** 500 delayed jobs: weather 28%, contractors 26%, parts 23%, crews 22%; median 9 extra days. No area >20 pts off field mix.	Mostly correct
Q11	**Hottest wellheads** Southeast Block 31 in 13 of 13 months; air temp spread ≤ 0.7 °C; holds within each well type.	Correct

Limits and next steps

- Q7 ranking and Q9.6 use a hand-designed score; the reasoning is shown, but it isn't validated.
- Keyword patterns for gas and abnormal-sound notes cover this dataset's phrasings; new phrasings need the pattern extended or the Gemma planner.
- Q10's “stands out” uses a fixed 20-point rule.
- Next: embedding retrieval over reports, multi-turn follow-ups, Gemma 4 spill-area estimation from real drone imagery.

Full answers, citations and visual data: `results/final.md` , `results/final.json` . Before the final-tools work, 0 of 21 turns were fully correct (2 partly).